What we found and what is next

Preliminary findings • Controlled, noise-free regression • GD vs L-BFGS

MAIN FINDING

As the valley narrows, GD slows sharply.

746

GD updates

5

L-BFGS iterations

At κ = 100 · Same start and gradient-norm threshold

NEXT STEPS

1 **Verify stopping criteria**

2 **Compare gradient evaluations and runtime**

3 **Test mini-batch SGD and feature scaling (if time)**

PRESENTATION PLAN

1:15 + 1:30 + 1:30 + 1:45 + 1:00 = 7:00

Rehearsal target: 6:30 • 30-second buffer

Results are preliminary; iteration counts are not a direct measure of total computational cost.